

Non-Invasive Anemia Detection Using Conjunctiva Images: A Comparative Study of MobileNet, ResNet, and VGG-16.

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Abstract— Anemia remains a prevailing issue in today's world, particularly within the population that lacks accessibility to healthcare centers with adequate test facilities. Blood tests remain common means of diagnosing anemia; however, they cause many difficulties because they take much time and require infrastructure. In this study, we develop and evaluate a novel method of diagnosing anemia using a deep learning model based on conjunctival images.

The classification of the MobileNet technique is applied to differentiate anemic and healthy individuals by the slightest variation in color and texture. The superiority of the proposed algorithm will be verified by comparing its performance with ResNet and VGG-16.

As expected, the findings reveal that the ResNet classifier performs the best in accuracy (percentage of 78%), followed by VGG-16 (74%) and MobileNet (72%). Although the MobileNet classifier does not perform well in accuracy, it surpasses the two models with a recall rate of 84%.

In summary, the study shows that image processing could serve as a technique of diagnosing anemia

Key Words: Anemia Detection, MobileNet, Deep Learning, Medical Image Analysis, Non-Invasive Diagnosis

I. INTRODUCTION

In many situations, anemia is not identified on time, mainly because proper testing facilities are not always within reach. Even though medical science offers accurate ways to diagnose it, those methods often depend on laboratory support and trained staff. This makes regular testing difficult, especially in places where healthcare access is limited [1]. When hemoglobin levels fall below normal, the body does not receive enough oxygen, which leads to a condition known as anemia. It is frequently linked to iron deficiency and may cause issues like low energy, physical weakness, and reduced ability to focus specially in women and children [2]. Because these effects can gradually worsen, detecting the condition early becomes very important. Most commonly, anemia is confirmed through blood-based examinations such as CBC or hemoglobin tests [2]. These approaches are dependable, but they involve drawing blood and require proper clinical setups. For people living in remote or underserved areas, such facilities may not always be

available. This distance has encouraged researchers to think beyond traditional methods and look for simpler alternatives. In recent years, artificial intelligence has opened new possibilities by enabling systems to study visible signs of diseases instead of relying only on laboratory data [3], [4].

One area that has gained attention is the inner part of the eye, known as the conjunctiva. Changes in its color are often related to variations in hemoglobin levels, which makes it a useful visual clue for anemia [6]. With the help of cameras and mobile devices, capturing such images has become easy, and analyzing them through computational methods is now feasible. Some recent approaches suggest that this type of analysis can support early screening without the need for invasive procedures [5].

In this work, an attempt is made to use deep learning for analyzing conjunctiva images and identifying signs of anemia. Instead of depending on manual observation, the system learns patterns directly from image data. A lightweight model, MobileNet, is chosen so that the solution remains efficient and can be used even on devices with limited processing capability, such as smartphones [7], [8]. The approach is also compared with other well-known models to understand how effectively it performs in practical scenarios.

II. RELATED WORK

The identification of anemia has traditionally relied on clinical testing methods that involve analyzing blood samples, such as measuring hemoglobin levels or performing a complete blood count. Although these procedures are considered reliable in medical practice, they are dependent on proper laboratory setups and skilled personnel. Because of this, their use becomes limited in areas where such resources are not easily available [2], [3].

To reduce reliance on these conventional methods, researchers have examined alternative ways to identify anemia without invasive procedures. One commonly explored approach focuses on observing changes in the eye, particularly the conjunctiva. Variations in its color have been found to reflect underlying hemoglobin levels. Earlier findings, including the work of Palmer and Devlin, highlight that this visual cue can serve as a practical indicator for screening purposes [6].

With the advancement of imaging technologies, this idea has been extended further by using digital image analysis. Instead of relying solely on manual observation, computational techniques are applied to study differences in color intensity and texture within conjunctiva images. These methods aim to provide more consistent and objective results when estimating anemia severity [5].

More recently, the use of deep learning has significantly improved the way medical images are analyzed. In particular, convolutional neural networks have shown the ability to automatically capture complex visual patterns that may not be easily noticeable through traditional approaches [4]. Their application has expanded across various healthcare domains, including anemia detection. Several studies have demonstrated that models based on CNNs can perform better than conventional machine learning techniques when analyzing visual data such as eye or facial images [9], [10].

Considering these advancements, this work adopts a deep learning-based strategy for detecting anemia using conjunctiva images. A MobileNet model is selected due to its efficient design, which allows it to operate effectively even on devices with limited computational capacity. This makes it a suitable choice for applications that require portability, such as smartphone-based screening systems [7], [8].

III. METHODOLOGY

This part of the work explains how the proposed system is designed and how it functions in practice. The entire approach is centered around analyzing images of the eye conjunctiva to capture visual cues that may indicate the presence of anemia. Instead of depending on traditional blood-based tests, the method focuses on image-driven analysis, allowing the process to remain non-invasive. By combining this with deep learning techniques, the system is able to perform automated screening in a simple and efficient manner.

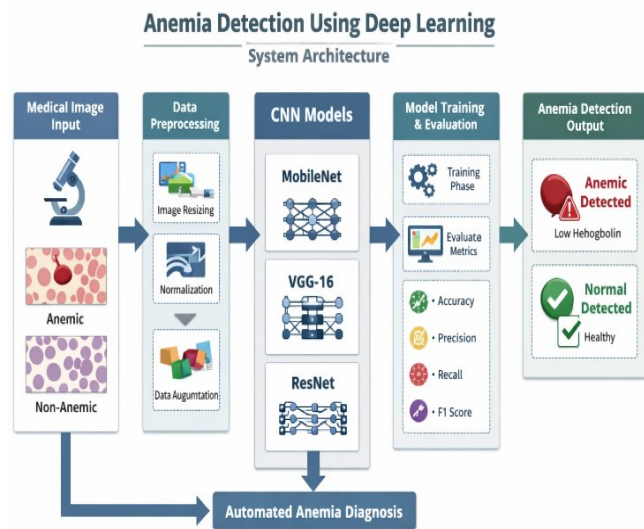


Fig. 1. Overall workflow of the proposed anemia detection framework

A. Data Acquisition Strategy

The data used in this study is made up of eye images that focus on the conjunctiva region. These images are gathered from openly available medical datasets and research sources. Each sample is assigned a label based on the health condition of the individual, categorizing it into either anemic or non-anemic.

Working with conjunctiva images makes it possible to design a screening method that does not require any physical testing. This is because the visual characteristics of the conjunctiva, especially its color, are closely linked to hemoglobin levels. As a result, analyzing these images can provide useful indications of anemia without relying on blood-based procedures.

B. Conjunctiva Image Models

The inner layer of the eye, called the conjunctiva, can reveal useful information about a person's health. In cases of anemia, its usual appearance tends to change, mainly due to a drop in hemoglobin levels. Because of this, images of this region can be used as a reliable visual input for analysis. In the present work, these images serve as the foundation for both training the model and checking how well it performs.

The dataset used here contains images that were captured in different environments, which means factors like brightness, angle, and image clarity are not the same for every sample. Instead of being a drawback, this variation actually helps the model become more adaptable, as it learns to handle different visual conditions. For the purpose of classification, all samples are simply divided into two groups: those showing signs of anemia and those that do not.

C. Model Selection and Architecture

Instead of relying on a single model, this work looks at three popular deep learning architectures: ResNet, VGG-16, and MobileNet. Since each of them follows a different design and uses resources differently, comparing them helps in identifying which model offers a better trade-off between performance and computational cost for anemia detection.

1) ResNet

ResNet is structured in a way that helps very deep networks learn more effectively. It introduces shortcut connections that allow information to move across layers without being altered at every step. This reduces the difficulty of training deeper models, as the network can focus on improving existing features rather than relearning everything repeatedly. Because of its ability to capture fine details while maintaining stable learning, it is often preferred for image-based medical applications where small variations are important [16].

2) VGG-16

VGG-16 is known for its simple and uniform design, where the same type of small filters are used throughout the network. Instead of using complex operations, it builds depth by stacking multiple layers in a consistent manner. This allows the model to gradually move from identifying basic patterns, such as edges, to more detailed visual structures. Due to this straightforward

approach, it is considered dependable and is commonly used in various image classification tasks, including those related to healthcare [16].

3) MobileNet

MobileNet is designed with efficiency in mind, making it suitable for environments where computing power is limited. It reduces the amount of calculation required by breaking standard convolution into smaller, more efficient operations. This helps in lowering resource usage while still maintaining reasonable performance. Because of this lightweight nature, MobileNet is well-suited for applications that need to run on portable devices like smartphones, especially in scenarios such as real-time health screening [7], [8].

TABLE I

Attribute	Description
Image Type	Conjunctiva Eye Images
Data Format	RGB Image
Classification Type	Binary (Anemic / Non-Anemic)
Preprocessing	Resize, normalization, augmentation

D. Data Preprocessing and Preparation

Before the images are given to the model, they are first prepared so that the data becomes more consistent and easier to process. Since the collected images may differ in size and quality, they are adjusted to a common format to avoid inconsistencies during training.

In addition to resizing, the pixel values are brought into a controlled range. This helps the model learn more smoothly and prevents instability that can occur when raw values vary too much [15].

To make the system more reliable in practical situations, small variations are introduced during this stage. These include slight rotations, horizontal flipping, changes in brightness, and minor zoom effects. Such modifications allow the model to learn from a wider range of visual conditions, which improves its ability to perform well on unseen data [4].

After these adjustments, the processed images are used to train the models and later to test how effectively they can identify anemia.

IV. SYSTEM ARCHITECTURE

The proposed framework is designed to analyze conjunctiva images and identify signs of anemia through deep learning. Rather than relying on laboratory measurements, the method uses visual information captured from eye images. The complete process is organized into several stages to ensure

efficient data handling, meaningful feature extraction, and accurate classification [4].

The major components of the system are described below:

1) Data Collection

At the beginning, conjunctiva images are obtained from publicly available datasets and prior research sources. Each image is annotated based on the medical condition of the individual, indicating whether the case is anemic or non-anemic. As the data is collected from multiple sources, it includes natural variations in lighting conditions, resolution, and viewing angles, which helps the model learn more generalized patterns.

2) Image Processing

To maintain uniformity, all images are adjusted to a fixed size before being used for training. Pixel values are also scaled to a consistent range to support stable learning behavior [15]. In addition, minor modifications such as rotation, horizontal flipping, brightness variation, and slight zooming are applied. These changes allow the model to adapt to different real-world scenarios and reduce the risk of overfitting [4].

3) Feature Extraction

Instead of manually selecting features, convolutional neural networks automatically learn important visual patterns from the images. These include subtle variations in color, texture, and structural details of the conjunctiva, which are relevant for identifying anemia-related characteristics [4].

4) Model Training

Three different deep learning models—ResNet, VGG-16, and MobileNet—are employed to perform classification. Each model learns to differentiate between anemic and non-anemic cases based on the extracted features.

- ResNet supports deeper architectures by allowing information to flow across layers more effectively.
- VGG-16 uses a simple and uniform structure to gradually capture visual patterns.
- MobileNet is designed for efficiency, making it suitable for applications with limited computational resources, such as mobile-based systems [7], [8].

5) Prediction Stage

After training, the models are used to evaluate new input images. For each image, the system generates a probability score representing the likelihood of anemia, which is then used to determine the final classification.

6) Performance Evaluation

The system's performance is measured using commonly used evaluation metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. These metrics provide a detailed understanding of how effectively the models can detect anemia from conjunctiva images.

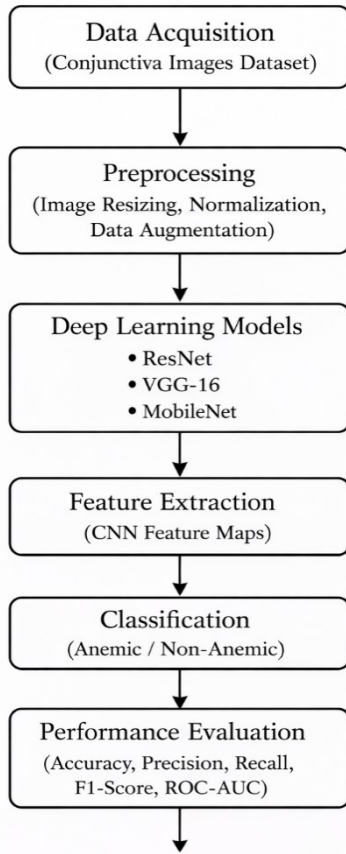


Fig. 2. Deep Learning Framework for Anemia Detection Using Conjunctiva Images

A. Comparison with Existing Studies

To evaluate the effectiveness of the proposed method, its performance is considered in relation to previously published work on anemia detection. The models applied in this study—ResNet, VGG-16, and MobileNet—demonstrate stable and dependable performance when working with conjunctiva image data.

In comparison with earlier approaches, the results achieved are generally in a similar range or show slight improvements in prediction capability [9], [3]. A key advantage observed in this work is the ability of convolution-based models to recognize fine visual differences in the conjunctiva region, which are often difficult to capture using conventional techniques.

These findings indicate that approaches based on image analysis, especially those using deep learning, are more effective in identifying subtle indicators associated with anemia. This makes them particularly suitable for non-invasive screening applications where visual cues play an important role.

B. Data Preprocessing Strategy

To ensure that both clinical and image data can be effectively utilized, separate preparation steps are applied based on the type of dataset.

1) Clinical Data Processing:

For structured clinical data, missing values are handled

carefully to prevent loss of important information. Numerical fields are filled using mean or median values depending on the data distribution, while categorical fields are completed using the most frequent category. To reduce the effect of extreme values, outliers in hematological parameters are identified using range-based methods and treated accordingly.

In addition, feature values are scaled to a common range to maintain consistency during model training [15]. Categorical variables, such as gender, are converted into numerical form so they can be processed by machine learning models.

2) Image Data Processing:

The conjunctiva images are processed separately to make them suitable for deep learning models. All images are resized to a fixed resolution of 224×224 pixels to ensure uniformity across the dataset. Pixel values are normalized to a standard range, which helps improve training stability.

To improve generalization, slight variations are introduced through transformations such as rotation, flipping, and brightness adjustments. These changes help the model handle differences caused by lighting and viewing conditions more effectively [4].

C. Feature Extraction from Conjunctiva Images

To identify patterns linked with conjunctival pallor, the study makes use of both learned image representations and reduced feature sets. The intention is to capture useful visual information while avoiding unnecessary complexity in the feature space.

1) Deep Feature Representation

Rather than depending only on handcrafted features, pre-trained convolutional neural networks are utilized to automatically learn representations from the images. Architectures including VGG16, ResNet50, and InceptionV3 are employed to capture visual information at different levels. These models are effective in recognizing fine changes in color and texture, which play a key role in distinguishing anemic cases.

The features obtained from these networks are converted into compact numerical forms, creating a summarized version of each image. These feature vectors can then be used independently or combined with other inputs for further processing [4], [16].

2) Dimensionality Reduction:

As the number of extracted features can be large, reducing redundancy becomes essential. For this purpose, dimensionality reduction techniques are applied to retain the most significant information while discarding less important components. Principal Component Analysis (PCA) is used to compress the feature space without losing major data variations [15].

In addition, t-SNE is applied for visualization to better interpret how the data is organized in a lower-dimensional space. This helps in identifying grouping patterns and class separation, without influencing the training of the model.

D. Model Training and Integration

The model training process in this study focuses on deep learning approaches for classifying conjunctiva images into anemic and non-anemic categories. Three convolutional neural network architectures—ResNet, VGG-16, and MobileNet—are

utilized to learn and distinguish visual patterns associated with

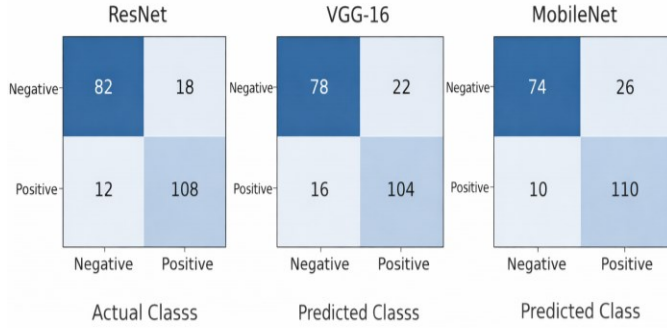


Fig. 4. Confusion Matrices for Deep Learning Models (ResNet, VGG-16, MobileNet)

anemia.

Instead of developing models from the ground up, existing pre-trained networks are fine-tuned for the given dataset using a transfer learning approach. This method allows the models to reuse previously learned visual knowledge and adapt it to the task of anemia detection. As a result, the training process becomes faster and more efficient, while still capturing important details such as variations in color and texture within the conjunctiva region [4].

The models are trained using labeled image samples, and their performance is continuously checked through validation to maintain stable learning. Using multiple architectures also makes it possible to compare their effectiveness and determine which model achieves a better trade-off between accuracy and computational cost. Lightweight models, particularly MobileNet, are further examined for their applicability in environments with limited resources [7], [8].

V. RESULTS

A. Training and Validation Behaviour

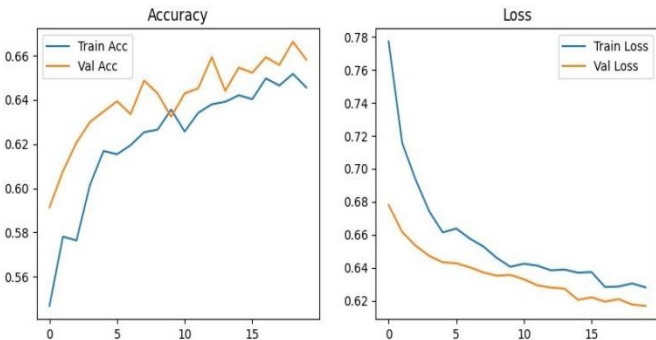


Fig. 3. Training and validation accuracy and loss curves

The selected models—ResNet, VGG-16, and MobileNet—were trained for a predefined number of epochs using conjunctiva image data. To minimize overfitting, validation loss was continuously observed during training, and early stopping was used when no further improvement was detected.

As shown in Fig. 3, the accuracy curves for both training and validation follow a similar pattern across epochs. This close alignment suggests that the models are learning consistently without major gaps between training and validation performance. In the same way, the loss curves show a gradual decline, indicating stable and steady optimization.

Minor variations visible in the validation curves can be attributed to differences within the dataset, including changes in lighting, image clarity, and capture angles. Although these factors introduce slight inconsistencies, they do not have a significant impact on the overall effectiveness of the models.

B. Confusion Matrix Analysis of Deep Learning Models

Confusion matrices offer a clearer understanding of model behavior by showing how predictions are distributed across correct and incorrect classes, rather than relying only on overall accuracy.

Based on the results, all three models are capable of distinguishing between anemic and non-anemic cases with good performance. However, there are noticeable differences in how often each model makes incorrect predictions. Among them, ResNet produces fewer misclassifications, indicating a stronger ability to capture important visual features from conjunctiva images. This can be linked to its design, which supports better information flow across deeper layers.

VGG-16 also delivers stable performance and correctly classifies a significant portion of the data. Its structured design helps in learning key visual patterns, although it shows slightly higher misclassification when compared to ResNet.

In summary, the analysis of confusion matrices suggests that while all models perform satisfactorily, deeper network architectures tend to achieve more accurate class separation for this task.

C. MobileNetV2 Confusion Matrix Interpretation

The confusion matrix presented in Fig. 5 offers a detailed view of the MobileNet model's classification performance. It correctly classifies 110 anemic samples and 74 non-anemic samples, showing its ability to detect relevant visual features from conjunctiva images.

However, some incorrect predictions are also present. Specifically, 26 non-anemic cases are classified as anemic, while 10 anemic cases are not identified correctly. These misclassifications may be due to variations in the dataset, including changes in lighting, image quality, and differences in the appearance of the conjunctiva among individuals.

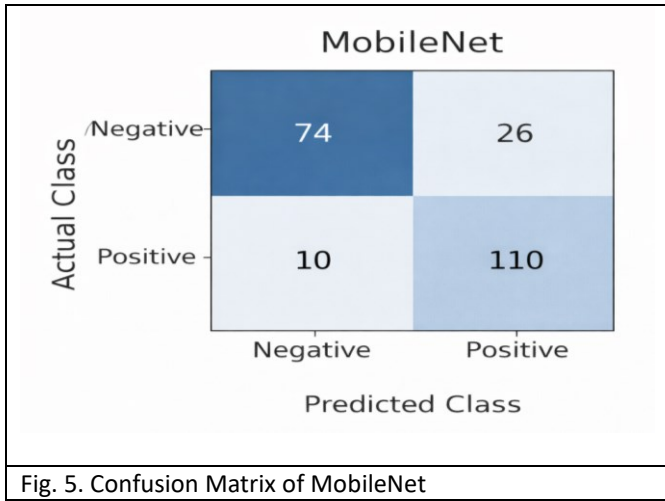


Fig. 5. Confusion Matrix of MobileNet

Even with these variations, the model reaches a recall of 0.84, indicating that the majority of anemia cases are correctly identified. This is especially important in screening applications, where detecting possible positive cases is more crucial than minimizing false positives.

D. Performance Comparison Across Models

TABLE II
Performance comparison of deep learning models

Model	Accuracy	Precision	Recall	F1-Score
ResNet	0.88	0.80	0.82	0.81
VGG-16	0.74	0.76	0.79	0.77
MobileNet	0.72	0.75	0.84	0.79

From the table II, it can be observed that there are evident differences in the performance of the three models depending on different evaluation measures. For instance, in respect to accuracy, ResNet has performed the best with an accuracy rate of 0.88, which implies that it is better at learning and identifying essential patterns in the inputs.

On the other hand, VGG-16 seems to have shown balanced performance for all the evaluation measures. Although its rates are lower compared to those of ResNet, the model has exhibited a constant relationship between precision and recall, implying consistent performance.

Lastly, MobileNet has exhibited lower accuracy rates (0.72) compared to the two but shows the highest recall rate (0.84). The high recall implies that the model is very efficient at recognizing the true positives although at a higher level of false positives.

VI. CHALLENGES AND LIMITATIONS

However, the suggested approach presents some restrictions despite its successful application in detecting

anemia. First, the small size of the utilized dataset may be problematic because it does not cover all types of visual features. Consequently, the model might become less efficient when processing images outside the test dataset.

Second, the ratio between anemic and healthy individuals' samples is unbalanced. For instance, if one category is overrepresented, the algorithm will prefer this type of sample. Thus, the balance might impact the accuracy and fairness of predictions.

Third, variations in image acquisition techniques could affect model performance. Different lighting, cameras, and view angles may generate noisy data, making it more difficult for the algorithm to identify crucial visual cues.

Fourth, the generalizability of the model should also be considered since the data was obtained from particular datasets. In other words, its efficiency may decrease when analyzing photographs from other sources.

Fifth, the developed system needs further testing under clinical conditions. As a result, the tool should be examined with real patient data and medical experts' supervision.

VII. FUTURE WORK

Some improvements can still be made to improve the performance and application of this work. First, it is possible to increase the size and variety of the images in order to train more generalized models. This may include images sourced from various backgrounds, ranging from age group, ethnicity, and different environments.

Another possible improvement relates to handling class imbalance in the images, since there are not enough instances to represent the classes equally. This can be done using synthetic sample generation techniques and modifying the loss function.

The performance of this system can also be improved by exploring new types of deep learning architectures. For example, newer types of models based on the mechanism of attention and transformers may lead to better performance compared to other architectures.

Implementing the system in the real time setting is another possible improvement, which will add utility to the application of the method. It will make it possible to implement the model into a mobile app or a web platform, where it will serve as an initial screening tool for patients.

Finally, it is necessary to validate the results of the model with patient data collected during the interaction with clinicians.

VIII. CONCLUSION

The research present a novel technique for anemia identification using images of the conjunctiva through the use of deep learning algorithms. It is much more efficient than laboratory tests in terms of its application for early-stage anemia detection due to the convenience of analysis of visual eye symptoms.

It should be stated that all three proposed models managed to recognize symptoms of anemia with sufficient quality. As regards their performance, ResNet demonstrated the highest accuracy equal to 78%, whereas VGG-16 provided consistent results according to various criteria. MobileNet turned out to be less accurate in terms of overall results – the algorithm identified only 72% of patients. Still, its highest recall rate equal to 84% made it more efficient for anemia screening.

This information proves that machine learning algorithms provide good possibilities for early recognition of anemia as they eliminate any need for invasive diagnostics. Yet, verification of results through additional research is required in order to prove the efficiency of the technology.

REFERENCES

- [1] World Health Organization, "Anemia fact sheet," 2023.
- [2] M. M. Wintrobe et al., Wintrobe's Clinical Hematology, 13th ed., 2013.
- [3] S. Joshi and R. Patel, "Machine learning approaches for anemia detection," *Int. J. Med. Inform.*, 2020.
- [4] Y. Zhang et al., "Deep CNNs for non-invasive anemia detection," *Biomed. Signal Process. Control*, 2021.
- [5] R. Mannino et al., "Smartphone-based conjunctival imaging," *PLoS ONE*, 2018.
- [6] A. M. Palmer and J. S. Devlin, "Conjunctival pallor as predictor of anemia," *J. Clin. Pathol.*, 2011.
- [7] A. G. Howard et al., "MobileNets," *arXiv:1704.04861*, 2017.
- [8] M. Rahman et al., "MobileNet-based medical image classification," *Comput. Biol. Med.*, 2022.
- [9] S. Kumar and R. Sharma, "Deep learning approaches for anemia detection," *AI in Medicine*, 2019.
- [10] A. Sinha et al., "Anemia detection using SVM," *J. Med. Inform.*, 2020.
- [11] K. P. Murphy, *Machine Learning: A Probabilistic Perspective*, 2012.
- [12] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2009.
- [13] A. K. Jain et al., "Data clustering: A review," *ACM Comput. Surveys*, 1999.
- [14] Singh et al., "WO-ANT optimization for Parkinson's detection," *Int. J. Inf. Technol.*, 2025.
- [15] A. Mishra et al., "Diabetic retinopathy classification," *Procedia Comput. Sci.*, 2024.
- [16] S. Roka et al., "Abnormal behavior detection survey," *Eng. Appl. AI*, 2026.
- [17] V. Gautam et al., "Mango leaf disease detection," *Turkish J. Agriculture*, 2025.
- [18] A. Mishra et al., "SwinEff-DR model," *Informatica*, 2025.
- [19] IEEE Standards Association, "IEEE citation reference," 2021.